IESA Semicon Hackathon

Idea Submission Template

Instructions

- Prepare the solution document as per this template, feel free to modify layouts while keeping core content .
- Kindly keep the maximum slides limit up to six (6-7). (Including the title slide)
- Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures .
- You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
- You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

FILE NAMING CONVENTION

Team Name_PSNo (E.g. i4C_PS01)

Team Details

Team Name: ScartchPad

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	**Team Leader**	*Gnanesh K*	*Fouth Year*
2	Member 1	*Baalekshan E*	*Fourth Year*
3	Member 2	*Anbu Saran G*	*Fourth Year*

A team can have up to 4 members including the team leader. Add rows if necessary.

COLLEGE NAME

Ramco Institute of Technology

TEAM LEADER CONTACT NUMBER

+91 8148717329

TEAM LEADER EMAIL ADDRESS

kgnanesh24@gmail.com

Problem Statement Addressed

Selected the problem statement your idea addresses

DESCRIPTION / DETAILS

The task is to locate a FinFET reference pattern within a larger SEM search image and return its center coordinate. It matters because repetitive nanoscale structures, noise, and distortion can cause navigation errors during semiconductor inspection.

Idea Description - Training-Free FinFET SEM Navigation Recovery

Provide a brief summary of your idea, including the key concept and approach. Provide a brief overview of your solution and how it addresses the problem statement.

KEY CONCEPT & APPROACH

Training-free classical image matching identifies the reference pattern in noisy FinFET SEM images using multi-scale correlation and center-rule refinement.

SOLUTION OVERVIEW

The system returns an accurate x,y location while handling repetitive structures, noise, blur, and small distortions without requiring a trained model.

Proposed Solution - Classical FinFET SEM Pattern Localization

Describe your idea in detail. Include the methodology, technologies involved, and how it addresses the chosen problem statement.

SOLUTION DETAILS

The solution uses Python, OpenCV, NumPy, and classical multi-scale template matching to compare a reference crop with the SEM search image. It evaluates candidate regions, applies center-rule refinement, and returns the best location as `x,y`, providing a lightweight, training-free method for recovering navigation errors in noisy, repetitive FinFET layouts.

Innovation and Uniqueness

Highlight what makes your idea unique or innovative compared to existing solutions.

KEY INNOVATION

A training-free FinFET SEM localizer combining multi-scale classical matching with center-rule refinement for reliable x,y recovery in repetitive patterns.

COMPETITIVE ADVANTAGE

Unlike ML-based approaches, it needs no training data, GPU, model download, or internet access, making it lightweight, reproducible, and easy to deploy.

Impact and Benefits

Explain how your solution will make an impact, such as improving performance, reducing costs, increasing efficiency, or solving other challenges.

Primary Impact

Reduces navigation-recovery time by automatically locating FinFET patterns, helping inspection workflows recover from image-position errors without trained AI models.

Quantifiable Outcomes

On the fixed synthetic FinFET evaluation pool: 80% accuracy within 5 px and 72.5% within 2 px; no GPU, model download, or network connection required.

Technology & Feasibility/Methodology Used

Describe the technologies, methodologies, or tools you plan to use to implement your idea.

IMPLEMENTATION STRATEGY

Python with OpenCV, NumPy, SciPy, and Streamlit. The system preprocesses grayscale SEM images, performs multi-scale classical matching, refines the center location, and returns x,y. It runs locally on a standard CPU—no GPU or trained model required.

Software Architecture

Hardware Components

Development Tools

GitHub & Video Link

GitHub Repository

https://github.com/GnaneshK24/ScratchPad/tree/main

Prototype / Simulation Video

https://github.com/GnaneshK24/ScratchPad/blob/main/docs/demo/finfet_hackathon_demo_edited.mp4

Research and References

Research Background & Methodology

Briefly describe the research foundation or scientific principles supporting your idea.

The method is based on classical image registration and template matching, supported by SEM principles such as noise, charging effects, and scan-related distortions. Multi-scale matching and correlation-based alignment are used to locate the reference pattern robustly in repetitive FinFET layouts.

References & Citations

List key papers, articles, or data sources.

OpenCV — Template Matching:
https://docs.opencv.org/4.12.0/de/da9/tutorial_template_matching.html

Evangelidis & Psarakis — Enhanced Correlation Coefficient Image Alignment (IEEE TPAMI, 2008):
https://doi.org/10.1109/TPAMI.2008.113

Joy — Noise in Secondary Electron Emission (Journal of Microscopy, 2005):
https://doi.org/10.1093/jmicro/dfi044